

# Homework

Note: sometimes your answer doesn't match one of the options exactly. That's fine. Select the option that's closest to your solution.

## Dataset

For this homework, we'll use the Car Fuel Efficiency dataset. Download it from [here](#).

You can do it with wget:

```
wget
https://raw.githubusercontent.com/alexeygrigorev/datasets/master/car_fuel_efficiency.csv
```

The goal of this homework is to create a regression model for predicting the car fuel efficiency (column `'fuel_efficiency_mpg'`).

## Preparing the dataset

Use only the following columns:

- `'engine_displacement'`,
- `'horsepower'`,
- `'vehicle_weight'`,
- `'model_year'`,
- `'fuel_efficiency_mpg'`

## EDA

- Look at the `'fuel_efficiency_mpg'` variable. Does it have a long tail?

## Question 1

There's one column with missing values. What is it?

- `'engine_displacement'`
- `'horsepower'`
- `'vehicle_weight'`
- `'model_year'`

## Question 2

What's the median (50% percentile) for variable `'horsepower'`?

- 49
- 99
- 149
- 199

## Prepare and split the dataset

- Shuffle the dataset (the filtered one you created above), use seed 42.
- Split your data in train/val/test sets, with 60%/20%/20% distribution.

Use the same code as in the lectures

## Question 3

- We need to deal with missing values for the column from Q1.
- We have two options: fill it with 0 or with the mean of this variable.
- Try both options. For each, train a linear regression model without regularization using the code from the lessons.
- For computing the mean, use the training only!
- Use the validation dataset to evaluate the models and compare the RMSE of each option.
- Round the RMSE scores to 2 decimal digits using `round(score, 2)`
- Which option gives better RMSE?

Options:

- With 0
- With mean
- Both are equally good

## Question 4

- Now let's train a regularized linear regression.
- For this question, fill the NAs with 0.
- Try different values of `r` from this list: `[0, 0.01, 0.1, 1, 5, 10, 100]`.
- Use RMSE to evaluate the model on the validation dataset.
- Round the RMSE scores to 2 decimal digits.
- Which `r` gives the best RMSE?

If multiple options give the same best RMSE, select the smallest `r`.

Options:

- 0
- 0.01
- 1
- 10
- 100

## Question 5

- We used seed 42 for splitting the data. Let's find out how selecting the seed influences our score.
- Try different seed values: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]`.
- For each seed, do the train/validation/test split with 60%/20%/20% distribution.
- Fill the missing values with 0 and train a model without regularization.
- For each seed, evaluate the model on the validation dataset and collect the RMSE scores.

- What's the standard deviation of all the scores? To compute the standard deviation, use `np.std`.
- Round the result to 3 decimal digits (`round(std, 3)`)

What's the value of std?

- 0.001
- 0.006
- 0.060
- 0.600

Note: Standard deviation shows how different the values are. If it's low, then all values are approximately the same. If it's high, the values are different. If standard deviation of scores is low, then our model is *stable*.

## Question 6

- Split the dataset like previously, use seed 9.
- Combine train and validation datasets.
- Fill the missing values with 0 and train a model with `r=0.001`.
- What's the RMSE on the test dataset?

Options:

- 0.15
- 0.515
- 5.15
- 51.5

## Submit the results

- Submit your results here: <https://courses.datatalks.club/ml-zoomcamp-2025/homework/hw02>
- If your answer doesn't match options exactly, select the closest one